Heng Zhang

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Dual-degree undergraduate in Computer Science and Business Analytics at NTU, with experience in MLP expert-head training and multimodal VAE modelling and validation. Conduct experiments with Python and PyTorch on remote GPUs, including data splitting, regularisation and latent-space comparisons, and evaluate models using ROC-AUC and C-index.

EDUCATION

Nanyang Technological University **Expected May 2028**

Dual degree: Bachelor of Computing (Computer Science) and Bachelor of Business (Business Analytics)

Relevant coursework: Artificial Intelligence (A-), Databases (A), Algorithm Design and Analysis (A-), Analytics I (A-)

RESEARCH EXPERIENCE

Nanyang Technological University | Generative AI Lab **Jul 2026 - Present**

Undergraduate Research Member | Supervisors: Prof. Dacheng Tao and Dr. Siyuan Liang (Postdoctoral Researcher)

Xidian University Hangzhou Research Institute **Feb 2026 - Present**

Research Assistant | Supervisor: Prof. Linhua Jiang

Binjiang Institute of Zhejiang University **Nov 2025 - Apr 2026**

Research Assistant | Supervisor: Dr. Meng Han

PROJECT EXPERIENCE

DUET AI-Generated Image Detection **2026**

Project: https://devpost.com/software/freedepth

TikTok TechJam 2026 | Expert-Head Training Contributor (five-person team)

- Trained MLP expert heads on frozen DINOv2 backbone features to detect AI-generated images, with training and validation targeting degradation from online sharing and compression.
- The team froze a 1.1-billion-parameter backbone and trained six MLP expert heads and a routing gate on a single NVIDIA RTX PRO 4000, with approximately 7 million trainable parameters.
- Training used 280,876 clean/degraded pairs with source-grouped data splits; performance was assessed on a separate set of 2,500 challenging validation images.
- Compared individual experts with fusion. The team fusion approach improved official-validation ROC-AUC by 3.39 percentage points over the best individual expert; early exit saved 16.4% of measured computation on average.

MO-RiskVAE | Binjiang Institute of Zhejiang University **Dec 2025 - Apr 2026**

Paper: https://arxiv.org/abs/2604.06267

- Contributed to a survival-supervised multimodal VAE integrating multi-omics fusion, missing-modality imputation and survival risk modelling; contributed to the manuscript as a co-first author.
- Compared KL, MMD and HSIC regularisation formulations and strengths under controlled training conditions, investigating the effects of regularisation and latent-space structure on performance.
- Proposed a continuous-discrete hybrid latent space using Gumbel-Softmax. The best single-run validation C-index reached 0.7788, an improvement of 0.0369 over the beta-KL baseline.
- Conducted evaluation using data from 1,143 patients and four external cohorts; contributed to results analysis, model-configuration comparisons and manuscript materials.

Meteorological AI Agent | Xidian Hangzhou Institute **Feb 2026 - Present**

Project: http://www.nextruments.com/

- Contribute to the experimental implementation of a knowledge-graph-enhanced weather agent combining meteorological observations, domain knowledge, graph retrieval and LLM reasoning.
- Design knowledge representations for weather variables, events and forecasting relationships, combining time-series observations with graph context for anomaly detection and trend analysis.
- Contribute to system design and experimental analysis, document methods and experiments, and analyse forecast explanations.

RAG Security and Robustness | ZJU Binjiang Institute **Nov 2025 - Apr 2026**

- Designed prompt-injection inputs and attack sets, using crawlers for RAG security and robustness testing and gaining experience in model-output validation.
- Compared model performance and analysed failure cases, documenting experimental results and examining how test inputs affect retrieval-augmented generation.
- Contributed to retrieval architecture and pipeline implementation exercises, organising application examples and technical teaching materials.

MOF-Agent | Binjiang Institute of Zhejiang University **Dec 2025 - Apr 2026**

- Implemented metal-organic framework representation and generation workflows, supported preliminary checks of structural validity and chemical feasibility, and contributed to manuscript preparation.

TECHNICAL SKILLS

Model training: Python, PyTorch, CUDA; MLP expert-head training on frozen backbone features; multimodal VAEs

Experimental design: Data preprocessing, source-grouped splits, validation-set evaluation, controlled regularisation and latent-space experiments

Models and metrics: DINOv2 features, RAG/GraphRAG, LLM agents; ROC-AUC, C-index

Computing and tools: Linux, SSH, remote GPU servers; Java, C/C++, Excel

Communication and documentation: Experimental analysis, technical writing, requirements analysis; Chinese and English (IELTS 7.0)

INTERNSHIP EXPERIENCE

China Merchants Bank **Aug 2025 - Oct 2025**

XinFuTong Enterprise FinTech Platform | Corporate Banking Intern

- Structured corporate client requirements, mapped business processes to digital financial-service workflows, and synthesised qualitative information into internal reporting documents.
- Organised client usage information and product feedback to support analysis of corporate banking processes and operational metrics.

CITIC Securities **May 2024 - Aug 2024**

Investment Banking Department | Part-time Assistant, Private-Placement Project Team

- Assisted in preparing private-placement offering documents and capital-markets presentation materials, consolidating industry analysis, fundraising plans and financial data.
- Researched compliance cases at peer securities firms, including CSC Financial, to support client-facing proposal materials.

PREPRINT

MO-RiskVAE: A Multi-Omics Variational Autoencoder for Survival Risk Modeling in Multiple Myeloma

Co-first author | arXiv preprint 2604.06267 (2026)